



NDVI Data Preparation Pipeline for California (July 2023)

Purpose

This note documents how I obtained and processed daily **NDVI (Normalized Difference Vegetation Index)** data for **California in July 2023** using open-source datasets from **NOAA Climate Data Record (CDR)**.

The resulting ndvi data can be used as a feature for **wildfire prediction**.



What is NDVI?

NDVI stands for **Normalized Difference Vegetation Index**, a measure of vegetation greenness.

It is calculated as:

$$NDVI = \frac{(NIR - RED)}{(NIR + RED)}$$

- **NIR**: Near-infrared reflectance (this value goes up when plants are healthy)
- **RED**: Red light reflectance (plants absorb for photosynthesis)

Value Range:

- $NDVI \in [-1, 1]$
- 0.6–1.0 → dense vegetation
- 0.2–0.5 → sparse/stressed vegetation
- ≤ 0 → water, bare soil, or clouds

Relation to wildfire:

Low NDVI → vegetation dryness → higher fire risk.

High NDVI → greener areas → less likely to burn immediately, but relevant for potential long-term fuel build-up.

Step 1. Obtain Global NDVI Data (July 2023)

Source: NOAA CDR (JP113C1) – <https://www.ncei.noaa.gov/data/land-normalized-difference-vegetation-index/access/2023/>

This directory contains **daily global NDVI files** in NetCDF format (.nc), each covering the entire Earth at 0.05° (~5 km) spatial resolution.

Automated download for 31 days in July 2023 using Python script `virris_download_auto.py`.

Result: 31 daily global NDVI .nc files stored in
`/raw_ndvi/viirs_daily/`



Step 2. Define California Boundary

Source: U.S. Census Bureau TIGER dataset (Cartographic Boundaries 2023).

File: `ca_boundary/cb_2023_us_state_20m.zip`

To limit NDVI data to California, I downloaded the California shapefile from U.S. Census Bureau TIGER dataset and created a **California boundary** using Python script `make_ca_boundary.py`.

Result: `ca_boundary/CA_boundary.shp` saved locally.



Step 3. Clip Global NDVI Data to California

Tools Used: `xarray` + `rioxarray` + `geopandas`

I used the Python script `ndvi_ca_polygon.py` to:

- open each daily global .nc file using `xarray`
- clip to the California polygon (via `rioxarray` + `GeoPandas`)
- apply the scale factor ($\text{NDVI} \times 0.0001$)
- and export results per day.

Output:

- 31 daily NDVI files:
`VIIRS-Land_v001_JP113C1_N0AA-20_202307XX_CApoly_lonlat_ndvi.parquet`
- Combined monthly file:
`JP113C1_CApoly_202307_all_days_lonlat_ndvi.parquet`

Columns: date , longitude , latitude , ndvi

Step 4. Export to CSV

Use Python script `ndvi_to_csv.py` to convert Parquet to CSV.

Final CSV:

`/raw_ndvi/JP113C1_CApoly_202307_all_days_lonlat_ndvi.csv`

Summary

Step	Description	Tools
1	Download 31 global NDVI .nc files	requests, bs4, tqdm
2	Obtain CA boundary	U.S. Census TIGER + GeoPandas
3	Clip to CA region	xarray, rioxarray
4	Extract lon/lat/ndvi	xarray → pandas
5	Save as Parquet/CSV	pandas, pyarrow

Output Overview

- **Source dataset:** NOAA/VIIRS JP113C1 NDVI (0.05° global grid)
- **Period:** July 1–31, 2023
- **Region:** California
- **Variables:** date , longitude , latitude , ndvi
- **Outputs:** 31 daily files + 1 combined monthly Parquet + 1 CSV